

Joint User and Target Tracking in Multi-static Sensing Systems

Boyang Hu*, Hui Tian*, Wei Ni^{†§}, and Shaoshuai Fan*

*State Key Laboratory of Networking and Switching Technology, Beijing University of Posts and Telecommunications, Beijing 100876, China

Email: {huboyang, tianhui, fanss}@bupt.edu.cn

[†]Data61, Commonwealth Scientific and Industrial Research Organisation, Sydney, NSW 2122, Australia

Email: wei.ni@data61.csiro.au

[§]School of Computer Science and Engineering, University of New South Wales, Sydney, NSW 2052, Australia

Abstract—Perceiving the mobility of the passive targets is critical in the context of Integrated Sensing and Communication (ISAC). While target localization has been extensively studied in radio sensing, the motion estimation and tracking of the targets have not been revealed. This paper proposes a new framework for joint UE and target tracking in the multi-static sensing system under the promising scenario that the user equipment (UE) serves as the receiver. We first estimate the UE's location using the direct paths between the UE and the base stations (BSs), and estimate the target's location of each scattering path to identify the targets by the mean-shift clustering algorithm. Then, we jointly estimate the motions of the UE and targets using the Taylor-least squares (LS) algorithm, and propose a joint UE and target tracking method based on the extended Kalman filter (EKF). Simulations demonstrate that the proposed joint motion estimation and tracking scheme improves the accuracy of location estimates by 42.34% w.r.t. the UE, and 71.29% w.r.t. the targets.

Index Terms—Multi-static sensing, target identification, motion estimation, joint user equipment (UE) and target tracking.

I. INTRODUCTION

Integrated Sensing and Communication (ISAC) is expected to be a key technique in the upcoming fifth-generation (5G) and sixth-generation (6G) wireless networks due to its potential for offering the communication systems the ability to perceive the surrounding environment [1]. The development of ISAC has spawned new high-quality connectivity- and high-accuracy sensing-based applications, such as vehicle to everything (V2X) and industrial internet of things (IoT) [2]. Instead of a dedicated and static base station (BS), using mobile user equipment (UEs) as the sensing receiver can save on infrastructure costs and, more importantly, enable range-unlimited and highly flexible sensing of targets.

As a key functionality of target sensing, target tracking demands the motion parameters, i.e., speeds and headings, of the targets to be estimated, which can only be inferred from the Doppler frequencies due to the passive nature of the targets. To unambiguously solve the speed and heading of a target, at least two paths from the target are needed, which requires target identification, i.e., associating the paths with the corresponding targets. However, existing literature on target tracking has not effectively addressed the problems of motion estimation and target identification.

On the one hand, the model of the Doppler frequency is simplified in many studies, limiting the implementations of the corresponding tracking methods. In [3], the targets were assumed to move with a known and fixed heading. In [4], [5], the Doppler frequency was simply modeled by the radial speed, where the motion estimation and tracking of the targets could not be achieved. Although [6], [7] correctly modeled the Doppler frequency (when the receiver was a BS), the target identities are assumed to be known *a priori*.

On the other hand, although target identification methods have been proposed in some studies, the motion estimation and tracking problems are still not well-addressed. In [3], a Kullback-Leiber divergence (KLD)-based target association method was proposed, where the motion parameters were assumed to be known *a priori*. In [8], the clustering method of density-based spatial clustering of applications with noise (DBSCAN) was used to identify multiple targets. However, the motion estimation and tracking problems were not mentioned.

In this paper, we solve the problems of target identification, motion estimation, and tracking in a unified framework, under a multi-static sensing scenario where the UE acts as the sensing receiver. In such cases, the problem is more complicated, since the motions and locations of the UE and the targets are tightly coupled in the measurements. To this end, we propose to jointly estimate the motion parameters and locations of the UE and the targets. Specifically,

- We estimate the UE's location using the direct paths between the UE and BSs. Then, we estimate the target location of each scattering path, and identify the targets using the mean-shift clustering algorithm.
- We jointly estimate the motion parameters of the UE and the targets from the Doppler frequencies of both the direct and scattering paths using an iterative Taylor-least squares (LS) estimator.
- We propose the extended Kalman filter (EKF) for joint UE and target tracking using both direct and scattering paths between the UE and BSs.

Simulations show that by jointly estimating the motions and locations of the UE and the targets, the tracking accuracy of the proposed scheme significantly outperforms the localization accuracy by up to 71.29%.

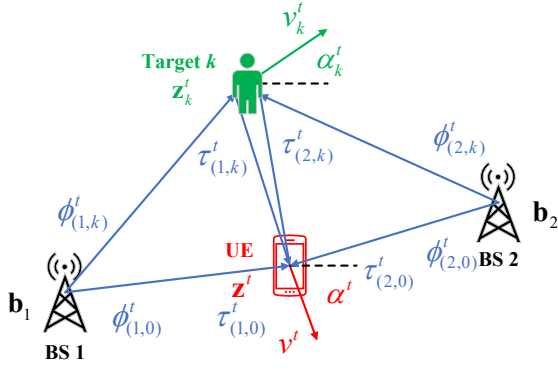


Fig. 1. Sensing scenario, where signals from the BSs are transmitted through direct (LoS) and scattering (NLoS) paths and received by the UE.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a two-dimensional (2D) multiple-input single-output (MISO) orthogonal frequency division multiplexing (OFDM) multi-static sensing scenario, as shown in Fig. 1, where there are I synchronized base stations (BSs) at locations $\mathbf{b}_i = [x_i^b, y_i^b]^T$, $\forall i = 1, \dots, I$, one UE situated at location $\mathbf{z}^t = [x^t, y^t]^T$ with speed v^t and heading α^t at the t -th measurement, and K sensing (passive) targets at locations $\mathbf{z}_k^t = [x_k^t, y_k^t]^T$ with speeds v_k^t and headings α_k^t , $\forall k = 1, \dots, K$. The clock offset between the UE and the BSs is δ^t . All the above states are unknown and to be estimated, except for I and $\mathbf{b}_i$, $\forall i$.

Definition 1 (Index of path). Let the (i, k) -th path denote the path transmitted from the i -th BS to the UE. When $k = 0$, the $(i, 0)$ -th path denotes the line-of-sight (LoS) path; When $k = 1, \dots, K$, the (i, k) -th path denotes the non-LoS (NLoS) path scattered on the k -th target and received by the UE.

We consider the measurement interval Δt as a short coherent processing interval (CPI), during which the measurements remain constant [9], [10]. Let $d_{i,k}^t, \phi_{i,k}^t, f_{i,k}^t$ denote the measured ToAs, AoDs, and Doppler frequencies of the (i, k) -th path, $k = 0, \dots, K$. At each downlink measurement, the ToAs, AoDs, and Doppler frequencies are jointly estimated and matched into three-dimensional (3D) tuples using the space-alternating expectation maximization (SAGE) algorithm [11]. When $k = 0$, the parameters of the $(i, 0)$ -th LoS path are given by

$$\text{ToA: } d_{i,0}^t = c(\tau_{i,0}^t + \delta^t) = \|\mathbf{b}_i - \mathbf{z}^t\|_2 + c\delta^t; \quad (1a)$$

$$\text{AoD: } \phi_{i,0}^t = \arctan2(y^t - y_i^b, x^t - x_i^b); \quad (1b)$$

$$\text{Doppler: } f_{i,0}^t = \frac{v^t}{\lambda} \left(\frac{x^t - x_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2} \cos \alpha^t + \frac{y^t - y_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2} \sin \alpha^t \right). \quad (1c)$$

When $k = 1, \dots, K$, the parameters of the (i, k) -th NLoS path are given by

$$\begin{aligned} \text{ToA: } d_{i,k}^t &= c(\tau_{i,k}^t + \delta^t) \\ &= \|\mathbf{b}_i - \mathbf{z}_k^t\|_2 + \|\mathbf{z}_k^t - \mathbf{z}^t\|_2 + c\delta^t; \end{aligned} \quad (2a)$$

$$\text{AoD: } \phi_{i,k}^t = \arctan2(y_k^t - y_i^b, x_k^t - x_i^b); \quad (2b)$$

$$\begin{aligned} \text{Doppler: } f_{i,k}^t &= \frac{1}{\lambda} \left\{ v^t \left(\frac{x^t - x_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \cos \alpha^t + \frac{y^t - y_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \sin \alpha^t \right) \right. \\ &\quad \left. + v_k^t \left[\left(\cos \phi_{i,k}^t - \frac{x^t - x_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \right) \cos \alpha_k^t + \left(\sin \phi_{i,k}^t - \frac{y^t - y_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \right) \sin \alpha_k^t \right] \right\}, \end{aligned} \quad (2c)$$

where c and λ are the propagation speed and wavelength of the signal, respectively, and $\arctan2(\cdot)$ is the four-quadrant inverse tangent. The derivations of the Doppler frequencies $f_{i,0}^t$ and $f_{i,k}^t$ are given in **Appendix A**.

We define the motion vectors of the UE and the k -th target as $\mathbf{v}^t = [v^t, \alpha^t]^T$ and $\mathbf{v}_k^t = [v_k^t, \alpha_k^t]^T$, respectively. Then, the state transition functions of $\mathbf{z}^t$ and $\mathbf{z}_k^t$, $\forall k$ are given by

$$\mathbf{z}^t = \mathbf{z}^{t-1} + v^{t-1} \Delta t [\cos \alpha^{t-1}, \sin \alpha^{t-1}]^T; \quad (3a)$$

$$\mathbf{z}_k^t = \mathbf{z}_k^{t-1} + v_k^{t-1} \Delta t [\cos \alpha_k^{t-1}, \sin \alpha_k^{t-1}]^T. \quad (3b)$$

Problem Statement: Given the measurements of ToAs, AoDs, and Doppler frequencies, we propose a framework for joint UE and target tracking. The motion parameters of the k -th target, i.e., $\mathbf{v}_k^t = [v_k^t, \alpha_k^t]^T$, can only be estimated using the Doppler frequencies from at least two NLoS paths (of the k -th target). Hence, target identification, which can associate the NLoS paths with the corresponding targets, is a prerequisite.

Hence, in the proposed framework, we first perform target identification and then estimate the targets' motions and track the targets' locations. Since the motions and locations of the UE and the targets are tightly coupled in the measurements, they are jointly estimated for improved accuracy.

III. PROPOSED FRAMEWORK

A. UE Localization

We estimate the location and clock offset of the UE using the ToAs and AoDs of the LoS paths.

Without loss of generality, we select the $(1, 0)$ -th LoS path as the reference path, and differentiate the ToAs into TDoAs. Then, we stack the TDoAs and AoDs of the LoS paths into the following matrices:

$$\begin{aligned} \mathbf{A} &= [\mathbf{a}_2^{\text{tdoa}}, \dots, \mathbf{a}_I^{\text{tdoa}}, \mathbf{a}_1^{\text{aod}}, \dots, \mathbf{a}_I^{\text{aod}}]^T \text{ with} \\ \mathbf{a}_i^{\text{tdoa}} &= [x_i^b - x_1^b, y_i^b - y_1^b, d_{i,0}^t - d_{1,0}^t]^T \text{ and} \\ \mathbf{a}_i^{\text{aod}} &= [-\sin \phi_{i,0}^t, \cos \phi_{i,0}^t, 0]^T; \\ \mathbf{c} &= [c_2^{\text{tdoa}}, \dots, c_I^{\text{tdoa}}, c_1^{\text{aod}}, \dots, c_I^{\text{aod}}]^T \text{ with} \\ c_i^{\text{tdoa}} &= \frac{1}{2} [\|\mathbf{b}_i\|_2^2 - \|\mathbf{b}_1\|_2^2 - (d_{i,0}^t - d_{1,0}^t)^2] \text{ and} \\ c_i^{\text{aod}} &= -x_i^b \sin \phi_{i,0}^t + y_i^b \cos \phi_{i,0}^t, \end{aligned} \quad (4)$$

where $\mathbf{A} \in \mathbb{R}^{(2I-1) \times 3}$. Define $\mathbf{z}_{\text{aux}}^t = [(\mathbf{z}^t)^T, \|\mathbf{z}^t - \mathbf{b}_1\|_2]^T$. Then, the location of the UE can be estimated using the following maximum likelihood estimator (MLE):

$$\tilde{\mathbf{z}}^t = \arg \min_{\mathbf{z}^t} (\mathbf{c} - \mathbf{A} \mathbf{z}_{\text{aux}}^t)^T \Phi^{-1} (\mathbf{c} - \mathbf{A} \mathbf{z}_{\text{aux}}^t), \quad (5)$$

where

$$\begin{aligned}\Phi &= \mathbf{E}\mathbf{Q}\mathbf{E}; \quad \mathbf{E} = \text{diag}(\mathbf{E}_{\text{tdoa}}, \mathbf{E}_{\text{aod}}) \text{ with} \\ \mathbf{E}_{\text{tdoa}} &= \text{diag}(e_2^{\text{tdoa}}, \dots, e_I^{\text{tdoa}}), \quad e_i^{\text{tdoa}} = \|\mathbf{z}^t - \mathbf{b}_i\|_2, \\ \mathbf{E}_{\text{aod}} &= \text{diag}(e_1^{\text{aod}}, \dots, e_I^{\text{aod}}) \text{ and} \\ e_i^{\text{aod}} &= \cos \phi_{i,0}^t (x^t - x_i^b) + \sin \phi_{i,0}^t (y^t - y_i^b); \\ \mathbf{Q} &= \text{diag}(\mathbf{Q}_{\text{tdoa}}, \mathbf{Q}_{\text{aod}}) \text{ with} \\ \mathbf{Q}_{\text{tdoa}} &= \text{diag}(\sigma_{d_{2,0}}^2, \dots, \sigma_{d_{I,0}}^2) + \sigma_{d_{1,0}}^2 \mathbf{1}_{(I-1) \times (I-1)} \text{ and} \\ \mathbf{Q}_{\text{aod}} &= \text{diag}(\sigma_{\phi_{1,0}}^2, \dots, \sigma_{\phi_{I,0}}^2).\end{aligned}$$

Here, $\mathbf{1}$ is a full-one matrix. $\mathbf{z}^t$ in e_i^{tdoa} and e_i^{aod} can be initialized by $\tilde{\mathbf{z}}^t = \arg \min (\mathbf{c} - \mathbf{A}\mathbf{z}_{\text{aux}}^t)^T (\mathbf{c} - \mathbf{A}\mathbf{z}_{\text{aux}}^t)$, which is the LS version of (5). (5) can be readily solved by Chan's algorithm [12].

Then, the clock offset can be estimated from the ToAs of the LoS paths, as given by

$$\tilde{\delta}^t = \sum_i (d_{i,0}^t - \|\mathbf{b}_i - \tilde{\mathbf{z}}^t\|_2) / (cI). \quad (6)$$

B. Target Identification

After the location $\tilde{\mathbf{z}}^t$ and clock offset $\tilde{\delta}^t$ of the UE are obtained, we proceed to estimate the target location w.r.t. each NLoS path and identify the targets using the mean-shift clustering algorithm. We also associate the targets in the time dimension for tracking.

The target location of the (i, k) -th NLoS path, denoted by $\mathbf{z}_{i,k}^t$, can be estimated by combining the ToA and AoD in (2):

$$\tilde{\mathbf{z}}_{i,k}^t = \mathbf{b}_i + \rho_{i,k} [\cos \phi_{i,k}^t, \sin \phi_{i,k}^t]^T, \quad (7)$$

where $\rho_{i,k}$ is the distance between the i -th BS and the k -th target, as given by $\rho_{i,k} = \frac{d_0^t - (x_i^b - \tilde{x}^t)^2 - (y_i^b - \tilde{y}^t)^2}{2[(x_i^b - \tilde{x}^t) \cos \phi_{i,k}^t + (y_i^b - \tilde{y}^t) \sin \phi_{i,k}^t + d_0^t]}$, with $d_0 = d_{i,k}^t - c\tilde{\delta}^t$.

In the multi-static sensing scenario, a target can be observed by multiple BSs, with its estimated locations $\tilde{\mathbf{z}}_{i,k}^t$, $\forall i$ situated around its real position. To this end, we use the mean-shift clustering algorithm [13] to find the NLoS paths associated with the same target, i.e., (i, k) , $\forall i$. Mean-shift clustering is a density-based non-parametric clustering method, which iteratively shifts the data points to the positions corresponding to the local maxima of the point density. Points converging to the same position belong to the same cluster. The mean-shift clustering algorithm is based on the kernel density estimate (KDE), where the most widely used kernel function is the Gaussian kernel. The only parameter that influences its performance is the kernel bandwidth, denoted by σ_{ms} , which can be reasonably adjusted according to the noise level. Interested readers may refer to [13] and [14] for more details.

Specifically, we input $\tilde{\mathbf{z}}_{i,k}^t$, $\forall (i, k)$ and obtain J clusters. We also obtain $\mathbb{C}_j$ and $\mathbf{c}_j = [x_j^c, y_j^c]^T$, $\forall j = 1, \dots, J$ as the path set and centroid of the j -th cluster, respectively. Without loss of generality, let cluster j associate with target k . Then, the state of the k -th target, i.e., $(\mathbf{z}_k^t, v_k^t, \alpha_k^t)$, can be accurately

estimated by combining its paths in $\mathbb{C}_j$, as will be discussed later in Sections III-C and III-D¹.

Then, the location of the k -th target can be estimated by averaging $\tilde{\mathbf{z}}_{i,k}^t$, $\forall i$, as given by

$$\tilde{\mathbf{z}}_k^t = \sum_i \tilde{\mathbf{z}}_{i,k}^t / I. \quad (8)$$

On the other hand, to keep track of the targets, we also need to associate the targets between consecutive measurements, i.e., associating target k_1 at time $t-1$ with target k_2 at time t . Let $\tilde{\mathbf{z}}_k^{t-1}$ denote the predicted location of the k -th target using (3b). If

$$\|\tilde{\mathbf{z}}_{k_1}^{t-1} - \mathbf{c}_{k_2}\|_2 \leq r_{\text{th}}, \quad (9)$$

then we determine that k_1 and k_2 correspond to the same target, where k_1 is a target at time $t-1$, and k_2 is a target at time t . For notation convenience, we use k_2 instead of j in $\mathbf{c}_{k_2}$. The threshold r_{th} can be selected based on the level of measurement noise.

C. Motion Estimation of UE and Targets

After the targets are identified, we can jointly estimate the motions of the UE, i.e., $\mathbf{v}^t = [v^t, \alpha^t]^T$ and the targets, i.e., $\mathbf{v}_k^t = [v_k^t, \alpha_k^t]^T$, $\forall k$ from the Doppler frequencies of the LoS and NLoS paths.

We define

$$\begin{aligned}\text{LoS: } x_{i,0}^g &= \frac{x^t - x_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2}, \quad y_{i,0}^g = \frac{y^t - y_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2}; \\ \text{NLoS: } x_{i,k}^g &= \frac{x^t - x_k^b}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2}, \quad y_{i,k}^g = \frac{y^t - y_k^b}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2},\end{aligned} \quad (10)$$

Then, we can rewrite (1c) and (2c) as

$$\begin{aligned}\text{LoS: } g_{i,0}(\mathbf{v}^t) &\triangleq v^t (x_{i,0}^g \cos \alpha^t + y_{i,0}^g \sin \alpha^t) = \lambda f_{i,0}^t; \\ \text{NLoS: } g_{i,k}(\mathbf{v}^t, \mathbf{v}_k^t) &\triangleq v^t (x_{i,k}^g \cos \alpha^t + y_{i,k}^g \sin \alpha^t) + \\ v_k^t [(\cos \phi_{i,k}^t - x_{i,k}^g) \cos \alpha_k^t + (\sin \phi_{i,k}^t - y_{i,k}^g) \sin \alpha_k^t] &= \lambda f_{i,k}^t.\end{aligned} \quad (11)$$

We then estimate $\mathbf{v}^t$ and $\mathbf{v}_k^t$ using the Taylor-LS algorithm [15]. Specifically, we linearize $g_{i,0}(\mathbf{v}^t)$ w.r.t. $\mathbf{v}^t$ and $g_{i,k}(\mathbf{v}^t, \mathbf{v}_k^t)$ w.r.t. $(\mathbf{v}^t, \mathbf{v}_k^t)$ by first-order Taylor expansion and solve (11) iteratively, as given by

$$\begin{aligned}\text{LoS: } g_{i,0}(\mathbf{v}^t) - g_{i,0}(\tilde{\mathbf{v}}^t) &\approx \frac{\partial g_{i,0}}{\partial \mathbf{v}^t}^T (\mathbf{v}^t - \tilde{\mathbf{v}}^t); \\ \text{NLoS: } g_{i,k}(\mathbf{v}^t, \mathbf{v}_k^t) - g_{i,k}(\tilde{\mathbf{v}}^t, \tilde{\mathbf{v}}_k^t) &= \\ \approx \frac{\partial g_{i,k}}{\partial \mathbf{v}^t}^T (\mathbf{v}^t - \tilde{\mathbf{v}}^t) + \frac{\partial g_{i,k}}{\partial \mathbf{v}_k^t}^T (\mathbf{v}_k^t - \tilde{\mathbf{v}}_k^t),\end{aligned} \quad (12)$$

where

$$\begin{aligned}\frac{\partial g_{i,k}}{\partial \mathbf{v}^t} &= \left[\frac{\partial g_{i,k}}{\partial v^t}, \frac{\partial g_{i,k}}{\partial \alpha^t} \right]^T \text{ with } \frac{\partial g_{i,k}}{\partial v^t} = x_{i,k}^g \cos \alpha^t + y_{i,k}^g \sin \alpha^t \\ \text{and } \frac{\partial g_{i,k}}{\partial \alpha^t} &= v^t (-x_{i,k}^g \sin \alpha^t + y_{i,k}^g \cos \alpha^t), \quad \forall k = 0, \dots, K; \\ \frac{\partial g_{i,k}}{\partial \mathbf{v}_k^t} &= \left[\frac{\partial g_{i,k}}{\partial v_k^t}, \frac{\partial g_{i,k}}{\partial \alpha_k^t} \right]^T, \quad \forall k = 1, \dots, K \text{ with} \\ \frac{\partial g_{i,k}}{\partial v_k^t} &= [(\cos \phi_{i,k}^t - x_{i,k}^g) \cos \alpha_k^t + (\sin \phi_{i,k}^t - y_{i,k}^g) \sin \alpha_k^t] \text{ and} \\ \frac{\partial g_{i,k}}{\partial \alpha_k^t} &= v_k^t [-(\cos \phi_{i,k}^t - x_{i,k}^g) \sin \alpha_k^t + (\sin \phi_{i,k}^t - y_{i,k}^g) \cos \alpha_k^t].\end{aligned}$$

¹One assumption of our target identification method is that the targets are not closely located. When some targets move close to each other (which can be inferred from the targets' motions), we can just keep locating them by (8), and restart to track their motions and locations after they move apart.

We define the joint motion vector

$$\mathbf{u}^t = [(\mathbf{v}^t)^T, (\mathbf{v}_1^t)^T, \dots, (\mathbf{v}_K^t)^T]^T \in \mathbb{R}^{2(K+1) \times 1}. \quad (13)$$

$\mathbf{u}^t$ can be estimated from (12) iteratively: In the j -th iteration, we rearrange (12) into matrix form by stacking the Doppler frequencies of all paths, as given by

$$\begin{aligned} \mathbf{f}_j &= [(\mathbf{f}_j^{\text{los}})^T, (\mathbf{f}_j^{\text{nlos}})^T]^T \in \mathbb{R}^{I(K+1) \times 1} \text{ with} \\ \mathbf{f}_j^{\text{los}} &= [\lambda f_{1,0}^t - g_{1,0}(\tilde{\mathbf{v}}^t), \dots, \lambda f_{I,0}^t - g_{I,0}(\tilde{\mathbf{v}}^t)]^T, \\ \mathbf{f}_j^{\text{nlos}} &= [(\mathbf{f}_{j,1}^{\text{nlos}})^T, \dots, (\mathbf{f}_{j,K}^{\text{nlos}})^T]^T \text{ and} \\ \mathbf{f}_{j,k}^{\text{nlos}} &= [\lambda f_{1,k}^t - g_{1,k}(\tilde{\mathbf{v}}^t, \tilde{\mathbf{v}}_k^t), \dots, \lambda f_{I,k}^t - g_{I,k}(\tilde{\mathbf{v}}^t, \tilde{\mathbf{v}}_k^t)]^T; \\ \mathbf{G}_j &= [(\mathbf{G}_j^{\text{los}})^T, (\mathbf{G}_j^{\text{nlos}})^T]^T \in \mathbb{R}^{I(K+1) \times 2(K+1)} \text{ with} \\ \mathbf{G}_j^{\text{los}} &= [\mathbf{G}_{j,\mathbf{v}}^{\text{los}}, \mathbf{0}_{I \times 2K}], \quad \mathbf{G}_{j,\mathbf{v}}^{\text{los}} = \left[\frac{\partial g_{1,0}}{\partial \mathbf{v}^t}, \dots, \frac{\partial g_{I,0}}{\partial \mathbf{v}^t} \right]^T, \\ \mathbf{G}_j^{\text{nlos}} &= [\mathbf{G}_{j,\mathbf{v}}^{\text{nlos}}, \text{diag}(\mathbf{G}_{j,\mathbf{v}_1}^{\text{nlos}}, \dots, \mathbf{G}_{j,\mathbf{v}_K}^{\text{nlos}})] \in \mathbb{R}^{IK \times 2(K+1)}, \\ \mathbf{G}_{j,\mathbf{v}}^{\text{nlos}} &= \left[\frac{\partial g_{1,1}}{\partial \mathbf{v}^t}, \dots, \frac{\partial g_{I,1}}{\partial \mathbf{v}^t}, \dots, \frac{\partial g_{1,K}}{\partial \mathbf{v}^t}, \dots, \frac{\partial g_{I,K}}{\partial \mathbf{v}^t} \right]^T \text{ and} \\ \mathbf{G}_{j,\mathbf{v}_k}^{\text{nlos}} &= \left[\frac{\partial g_{1,k}}{\partial \mathbf{v}^t}, \dots, \frac{\partial g_{I,k}}{\partial \mathbf{v}^t} \right]^T; \quad \mathbf{w}_j = \mathbf{u}^t - \tilde{\mathbf{u}}^t. \end{aligned}$$

Then, $\mathbf{w}_j$ can be estimated using the following MLE:

$$\tilde{\mathbf{w}}_j = \arg \min_{\mathbf{w}_j} (\mathbf{f}_j - \mathbf{G}_j \mathbf{w}_j)^T (\mathbf{f}_j - \mathbf{G}_j \mathbf{w}_j) = (\mathbf{G}_j^T \mathbf{G}_j)^{-1} \mathbf{G}_j^T \mathbf{f}_j. \quad (15)$$

Finally, $\mathbf{u}^t$ is updated iteratively until convergence:

$$(\tilde{\mathbf{u}}^t)_{j+1} = (\tilde{\mathbf{u}}^t)_j + \tilde{\mathbf{w}}_j, \quad (16)$$

where $(\tilde{\mathbf{u}}^t)_j$ is the estimate of $\mathbf{u}^t$ in the j -th iteration².

D. Joint UE and Target Tracking

Given the estimated clock offset $\tilde{\delta}^t$ and the motion parameters of the UE $\tilde{\mathbf{v}}^t$ and the targets $\tilde{\mathbf{v}}_k^t, \forall k$, we track the UE and the targets using the ToAs and AoDs. Due to the fact that the locations of the UE and the targets are highly coupled in the measurement model given by (2), we track the UE and the targets jointly using the EKF.

We first define the joint location vector as

$$\mathbf{z}_{\text{all}}^t = [(\mathbf{z}^t)^T, (\mathbf{z}_1^t)^T, \dots, (\mathbf{z}_K^t)^T]^T \in \mathbb{R}^{2(K+1) \times 1}. \quad (17)$$

We rewrite the state evolution model in (3) (denoted by $\ell(\cdot)$) and the measurement model in (1) and (2) (denoted by $\mathbf{h}(\cdot)$) as

$$\mathbf{z}_{\text{all}}^t = \ell(\mathbf{z}_{\text{all}}^{t-1}) + \mathbf{n}_\ell; \quad \mathbf{s}^t = \mathbf{h}(\mathbf{z}_{\text{all}}^t) + \mathbf{n}_h, \quad (18)$$

²The initial point of this iterative algorithm, i.e., $(\tilde{\mathbf{u}}^t)_1$, can be provided by $\tilde{\mathbf{u}}^{t-1}$, considering that the motions of the targets do not change much after a short time interval Δt , i.e., $\mathbf{u}^{t-1}$ is relatively close to $\mathbf{u}^t$. While $(\tilde{\mathbf{u}}^t)_1$ can be initialized as the center of the possible region of $\mathbf{u}$, e.g., $(\tilde{\mathbf{v}}^t)_1 = [0.5 \cdot v_{\max} \text{ m/s}, 0 \text{ rad}]^T$. This initialization method is demonstrated to be effective through simulations in Section IV; see Figs. 4 and 5, indicating that the Taylor-LS algorithm is not sensitive to the choice of initial points.

where

$$\begin{aligned} \mathbf{s}^t &= [(\mathbf{s}_{\text{los}}^t)^T, (\mathbf{s}_{\text{nlos}}^t)^T]^T, \\ \mathbf{s}_{\text{los}}^t &= [(\mathbf{s}_{1,0}^t)^T, \dots, (\mathbf{s}_{I,0}^t)^T]^T \text{ with } \mathbf{s}_{i,0}^t = [d_{i,0}^t, \phi_{i,0}^t]^T; \\ \mathbf{s}_{\text{nlos}}^t &= [(\mathbf{s}_{\text{nlos},1}^t)^T, \dots, (\mathbf{s}_{\text{nlos},K}^t)^T]^T \text{ with} \\ \mathbf{s}_{\text{nlos},k}^t &= [(\mathbf{s}_{1,k}^t)^T, \dots, (\mathbf{s}_{I,k}^t)^T]^T \text{ and } \mathbf{s}_{i,k}^t = [d_{i,k}^t, \phi_{i,k}^t]^T; \\ \mathbf{n}_\ell &= [\Delta \mathbf{z}_\ell^T, \Delta \mathbf{z}_{\ell,1}^T, \dots, \Delta \mathbf{z}_{\ell,K}^T]^T; \quad \mathbf{n}_h = \Delta \mathbf{s}^t. \end{aligned}$$

Here, $\mathbf{s}^t$ is the measurement vector that contains the ToAs and AoDs of LoS and NLoS paths; $\mathbf{n}_\ell$ and $\mathbf{n}_h$ are the zero-mean uncorrelated Gaussian motion disturbances [16] and measurement noises, respectively, with covariance matrices

$$\begin{aligned} \mathbf{Q}_\ell &= \text{diag}(\sigma_{x_\ell}^2, \sigma_{y_\ell}^2, \sigma_{x_{\ell,1}}^2, \sigma_{y_{\ell,1}}^2, \dots, \sigma_{x_{\ell,K}}^2, \sigma_{y_{\ell,K}}^2); \\ \mathbf{Q}_h &= \text{diag}(\sigma_{d_{i,0}^t}^2, \sigma_{\phi_{i,0}^t}^2, \sigma_{d_{i,1}^t}^2, \sigma_{\phi_{i,1}^t}^2, \dots, \sigma_{d_{i,K}^t}^2, \sigma_{\phi_{i,K}^t}^2). \end{aligned} \quad (19)$$

To estimate $\mathbf{z}_{\text{all}}^t$, the time update of the EKF is given by

$$\tilde{\mathbf{z}}_{\text{all}}^{t|t-1} = \ell(\tilde{\mathbf{z}}_{\text{all}}^{t-1}); \quad \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^{t|t-1} = \mathbf{L} \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^{t-1} \mathbf{L}^T + \mathbf{Q}_\ell, \quad (20)$$

where $\mathbf{L} = \frac{\partial \mathbf{z}_{\text{all}}^t}{\partial (\mathbf{z}_{\text{all}}^{t-1})^T} \stackrel{(3)}{=} \mathbf{I}_{2(K+1) \times 2(K+1)}$.

The measurement update of the EKF is given by

$$\begin{aligned} \tilde{\mathbf{z}}_{\text{all}}^t &= \tilde{\mathbf{z}}_{\text{all}}^{t|t-1} + \mathbf{W} [\mathbf{s}^t - \mathbf{h}(\tilde{\mathbf{z}}_{\text{all}}^{t|t-1})]; \\ \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^t &= [\mathbf{I}_{2(K+1) \times 2(K+1)} - \mathbf{W} \mathbf{H}] \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^{t|t-1}, \end{aligned} \quad (21)$$

where

$$\begin{aligned} \mathbf{W} &= \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^{t|t-1} \mathbf{H}^T (\mathbf{H} \mathbf{Q}_{\tilde{\mathbf{z}}_{\text{all}}}^{t|t-1} \mathbf{H}^T + \mathbf{Q}_h)^{-1}; \\ \mathbf{H} &= \frac{\partial \mathbf{s}^t}{\partial (\mathbf{z}_{\text{all}}^t)^T} = [(\mathbf{H}^{\text{los}})^T, (\mathbf{H}^{\text{nlos}})^T]^T \in \mathbb{R}^{2I(K+1) \times 2(K+1)}; \\ \mathbf{H}^{\text{los}} &= [\mathbf{H}_{\mathbf{z}}^{\text{los}}, \mathbf{0}_{2I \times 2K}]; \quad \mathbf{H}_{\mathbf{z}}^{\text{los}} = \frac{\partial \mathbf{s}_{\text{los}}^t}{\partial (\mathbf{z}^t)^T} \text{ with} \\ \frac{\partial d_{i,0}^t}{\partial (\mathbf{z}^t)^T} &= \frac{(\mathbf{z}^t - \mathbf{b}_i)^T}{\|\mathbf{z}^t - \mathbf{b}_i\|_2} \text{ and } \frac{\partial \phi_{i,0}^t}{\partial (\mathbf{z}^t)^T} = \left[\frac{y_i^b - y_i^t}{\|\mathbf{z}^t - \mathbf{b}_i\|_2^2}, \frac{x_i^t - x_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2^2} \right]; \\ \mathbf{H}^{\text{nlos}} &= [\mathbf{H}_{\mathbf{z}_1}^{\text{nlos}}, \text{diag}(\mathbf{H}_{\mathbf{z}_1}^{\text{nlos}}, \dots, \mathbf{H}_{\mathbf{z}_K}^{\text{nlos}})] \in \mathbb{R}^{2IK \times 2(K+1)}; \\ \mathbf{H}_{\mathbf{z}}^{\text{nlos}} &= \frac{\partial \mathbf{s}_{\text{nlos}}^t}{\partial (\mathbf{z}^t)^T} \in \mathbb{R}^{2IK \times 2} \text{ with} \\ \frac{\partial d_{i,k}^t}{\partial (\mathbf{z}^t)^T} &= \frac{(\mathbf{z}^t - \mathbf{z}_k^t)^T}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \text{ and } \frac{\partial \phi_{i,k}^t}{\partial (\mathbf{z}^t)^T} = [0, 0]; \\ \mathbf{H}_{\mathbf{z}_k}^{\text{nlos}} &= \frac{\partial \mathbf{s}_{\text{nlos},k}^t}{\partial (\mathbf{z}_k^t)^T} \in \mathbb{R}^{3I \times 2} \text{ with} \\ \frac{\partial d_{i,k}^t}{\partial (\mathbf{z}_k^t)^T} &= \frac{(\mathbf{z}_k^t - \mathbf{b}_i)^T}{\|\mathbf{b}_i - \mathbf{z}_k^t\|_2} + \frac{(\mathbf{z}_k^t - \mathbf{z}^t)^T}{\|\mathbf{z}_k^t - \mathbf{z}^t\|_2}, \quad \frac{\partial \phi_{i,k}^t}{\partial (\mathbf{z}_k^t)^T} = \left[\frac{y_i^b - y_k^t}{\|\mathbf{z}_k^t - \mathbf{b}_i\|_2^2}, \frac{x_k^t - x_i^b}{\|\mathbf{z}_k^t - \mathbf{b}_i\|_2^2} \right]. \end{aligned}$$

IV. SIMULATION RESULTS

In this section, we present simulation results to evaluate the sensing performance of the proposed scheme. The UE and $K = 3$ targets move within the region of $50 \text{ m} \times 40 \text{ m}$, where $I = 6$ BSs are located at $(0, 0)$, $(25, 0)$, $(50, 0)$, $(0, 40)$, $(25, 40)$, and $(50, 40)$, respectively. At each instant t , the speeds and headings of the UE and the targets are randomly generated within $[0, 10] \text{ m/s}$ and $[-\pi, \pi] \text{ rad}$, respectively, and the clock offsets are randomly generated within $[0, 300] \text{ ns}$. The measurement interval Δt is set as 100 ms . The standard deviations of the measurement errors

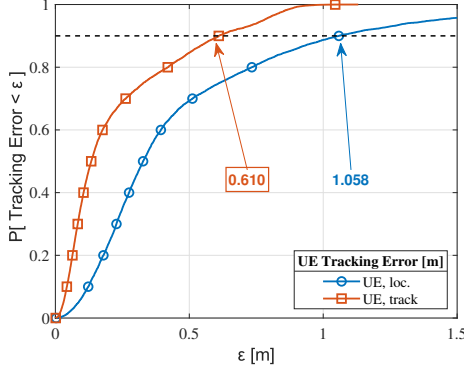


Fig. 2. The CDFs of the UE's localization and tracking errors.

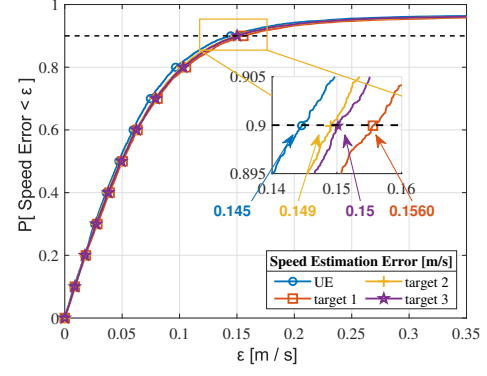


Fig. 4. The CDFs of the speed estimation errors.

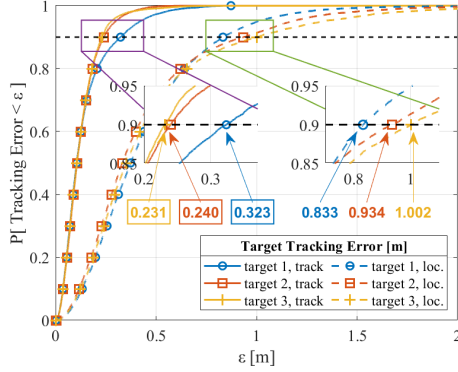


Fig. 3. The CDFs of the targets' localization and tracking errors.

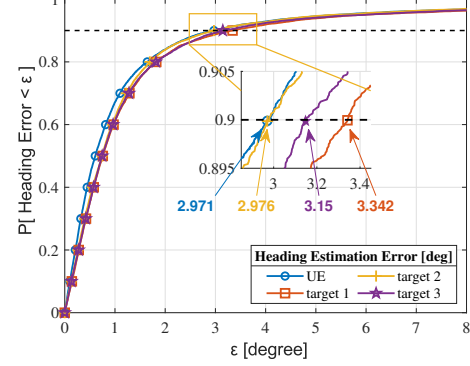


Fig. 5. The CDFs of the heading estimation errors.

of ToAs, AoDs, and Doppler frequencies are $\sigma_d = 0.1$ m, $\sigma_\phi = 1^\circ$, $\sigma_f = 10$ Hz. The motion disturbance error in (18) is set as $\mathbf{Q}_\ell = \sigma_\ell^2 \mathbf{I}_{2(K+1) \times 2(K+1)}$ with $\sigma_\ell = 0.1$ m.

Fig. 2 plots the cumulative distribution functions (CDFs) of the UE's localization and tracking errors. It is observed that the UE's localization and tracking accuracy of the proposed scheme achieves 1.058 m and 0.610 m, respectively. Compared to UE localization, which only uses the LoS paths, UE tracking uses the estimated motion parameters and the NLoS paths in the joint EKF algorithm and thus presents better performance (by 42.34%).

Fig. 3 provides the CDFs of the targets' localization and tracking errors. It is observed that all targets present similar accuracy in both localization (0.923 m on average) and tracking (0.265 m on average). Similar to the UE, a significant improvement (by 71.29%) from target localization to target tracking is observed, which is mainly contributed by the motion parameters and jointly estimating the targets' locations with the UE's location.

Figs. 4 and 5 depict the CDFs of the motion estimation errors. The speed estimation accuracy of the UE and the targets is 0.145 m/s and 0.152 m/s (in average), respectively, and the heading estimation accuracy of the UE and the targets is 2.971 degrees and 3.156 degrees (in average), respectively. As a result, the precisely estimated motion parameters of the proposed scheme are the basis of the high-accuracy UE and target tracking in Figs. 2 and 3.

V. CONCLUSION

In this paper, we studied the target tracking problem in the multi-static sensing scenario where the UE served as the receiver. To combine the measurements of the same target from multiple BSs, we proposed associating the NLoS paths with the targets using the mean-shift clustering algorithm. Then, to estimate the tightly coupled motions and locations of the UE and the targets from the measurements, we presented joint motion estimation and tracking methods based on the Taylor-LS and EKF algorithms, respectively. Simulations demonstrated that the proposed scheme could improve the location estimation accuracy by up to 71.29%.

VI. ACKNOWLEDGEMENT

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APPENDIX A

DERIVATIONS OF $f_{i,0}^t$ AND $f_{i,k}^t$ IN (1c) AND (2c)

We start from the Doppler frequency of the $(i,0)$ -th LoS path, i.e., $f_{i,0}^t$ in (1c). The change in the phase is caused by the change in the path length, i.e., $\Delta\varphi_{i,0}^t = \frac{2\pi\Delta d_{i,0}^t}{\lambda}$. Let $\Delta\mathbf{z}^t$ denote the location change of the UE during the measurement interval Δt . The first-order Taylor expansion of (1a) gives the approximation of $\Delta d_{i,0}^t$ as

$$\Delta d_{i,0}^t \approx \left(\frac{\partial d_{i,0}^t}{\partial \mathbf{z}^t} \right)^T \Delta \mathbf{z}^t, \quad (22)$$

where

$$\frac{\partial d_{i,0}^t}{\partial \mathbf{z}^t} = \left[\frac{x^t - x_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2}, \frac{y^t - y_i^b}{\|\mathbf{z}^t - \mathbf{b}_i\|_2} \right]^T; \Delta \mathbf{z}^t \stackrel{(3a)}{=} v^t \Delta t [\cos \alpha^t, \sin \alpha^t]^T.$$

Substituting (22) into $f_{i,0}^t = \frac{\Delta \varphi_{i,0}^t}{2\pi \Delta t} = \frac{\Delta d_{i,0}^t}{\lambda \Delta t}$ gives (1c).

We then proceed to the Doppler frequency of the (i, k) -th NLoS path, i.e., $f_{i,k}^t$ in (2c). Let $\Delta \mathbf{z}_k^t$ denote the location changes of the k -th target during the measurement interval Δt . The first-order Taylor expansion of (2a) gives the approximation of $\Delta d_{i,k}^t$ as

$$\Delta d_{i,k}^t \approx \left(\frac{\partial d_{i,k}^t}{\partial \mathbf{z}^t} \right)^T \Delta \mathbf{z}^t + \left(\frac{\partial d_{i,k}^t}{\partial \mathbf{z}_k^t} \right)^T \Delta \mathbf{z}_k^t, \quad (23)$$

where

$$\begin{aligned} \frac{\partial d_{i,k}^t}{\partial \mathbf{z}^t} &= \left[\frac{x^t - x_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2}, \frac{y^t - y_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \right]^T; \\ \frac{\partial d_{i,k}^t}{\partial \mathbf{z}_k^t} &= \left[\frac{x_k^t - x_i^b}{\|\mathbf{b}_i - \mathbf{z}_k^t\|_2} - \frac{x^t - x_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2}, \frac{y_k^t - y_i^b}{\|\mathbf{b}_i - \mathbf{z}_k^t\|_2} - \frac{y^t - y_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \right]^T \\ &\stackrel{(2b)}{=} \left[\cos \phi_{i,k}^t - \frac{x^t - x_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2}, \sin \phi_{i,k}^t - \frac{y^t - y_k^t}{\|\mathbf{z}^t - \mathbf{z}_k^t\|_2} \right]^T; \\ \Delta \mathbf{z}_k^t &\stackrel{(3b)}{=} v_k^t \Delta t [\cos \alpha_k^t, \sin \alpha_k^t]^T. \end{aligned}$$

Substituting (23) into $f_{i,k}^t = \frac{\Delta \varphi_{i,k}^t}{2\pi \Delta t} = \frac{\Delta d_{i,k}^t}{\lambda \Delta t}$ gives (2c).

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